10

Data, tables and graphs

Part V · Data and decisions

CHAPTER 10 · OPENER

Data, tables and graphs

Data displays and summaries make patterns visible, but choices of grouping, scale and denominator shape the claims that seem reasonable.

ENGAGEMENT

How can the same survey support a fair decision without letting a misleading graph or an unstated denominator do the arguing?

Learning goals

- Classify variables and organise frequency or two-way tables.
- Construct and interpret bar and pie charts.
- Summarise distributions with dot plots, histograms and averages.
- Describe trends and detect misleading scales or claims.

MYP lens

Key concept: Relationships

Related concepts: Representation, Pattern

Global context: Fairness and development

ATL focus. In order for a student to process and communicate information, students must organise data, choose an appropriate representation, calculate summaries and justify a claim with its limitations.

Language bank: variable · categorical · nominal · ordinal · discrete · continuous · frequency · two-way table · bar chart · pie chart · sector angle · dot plot · histogram · mean · median · mode · range · outlier · trend

CHAPTER 10 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	Find 96+74+30.	READY
R2	Find 96/200 as a percentage.	READY
R3	Find the mean of 2, 4, 6.	READY
R4	Find 360×0.25.	READY
R5	Order 5, 2, 8, 3.	READY

Chapter map

1. **10.1 Data, variables and tables** — Identify what was measured, the unit, the population and the type of variable before summarising.
2. **10.2 Bar charts and pie charts** — Use bars to compare categories and sectors to show parts of one whole.
3. **10.3 Dot plots, histograms and averages** — Describe shape, centre and spread; choose an average that fits the data and the question.
4. **10.4 Trends, comparisons and misleading graphs** — Read the scale, denominator and time order before accepting a visual claim.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 10 · MATHEMATICAL DEVELOPMENT

10.1 Data, variables and tables

BIG IDEA

Identify what was measured, the unit, the population and the type of variable before summarising.

Data are recorded values. A categorical variable names groups (nominal) or ordered categories (ordinal). A numerical variable is discrete if it counts separate values, or continuous if it measures on a scale.

A frequency table lists categories or intervals and their counts. Frequencies must sum to the number of observations. A two-way table shows how two categories overlap.

Frequency table

Lunch choice	Frequency
School	96
Packed	74
Off-site	30
Total	**200**

A frequency table organises categories; the counts sum to the 200 students surveyed.

WORKED EXAMPLE · CLASSIFY VARIABLES

Lunch choice, year group, number of siblings and height.

1. Lunch choice: nominal categorical.
2. Year group: ordinal categorical (or discrete numerical code).
3. Siblings: discrete numerical. Height: continuous numerical.

The variable type guides the display and summary.

CHAPTER 10 · PRACTICE

10.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A number used as a label (such as a shirt size code) is not automatically a quantity to average.

GUIDED PRACTICE · FIRST STEPS

10.1.1 Classify: favourite fruit, school year, number of pets, height. FLUENCY

10.1.2 A table has frequencies 38, 31, 27 for school lunch in Years 7–9. Find the total. FLUENCY

10.1.3 Why is “year group” not usually averaged as if it were height? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.1.4 In a two-way table, Year 7 has 38 school, 24 packed, 8 off-site. Find the row total and the school-lunch share of Year 7. APPLICATION

10.1.5 A survey records “0, 1, 2, 3” as codes for four lunch choices. Can these codes be added meaningfully? REASONING

10.1.6 A frequency table has total 200 and known frequencies 96 and 74. Find the missing frequency and its percentage. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · MATHEMATICAL DEVELOPMENT

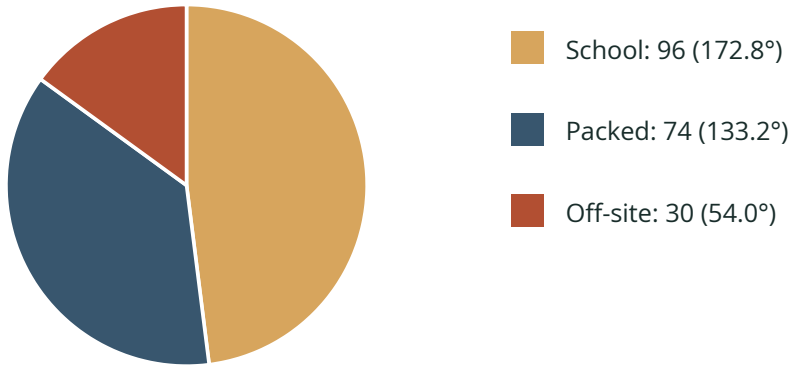
10.2 Bar charts and pie charts

BIG IDEA

Use bars to compare categories and sectors to show parts of one whole.

A bar chart uses equal-width bars with heights proportional to frequency. Label axes, start the frequency axis at zero and keep spacing consistent.

A pie chart represents a whole by 360°. Sector angle = category frequency/total × 360°. Percentages should sum to 100% apart from stated rounding.



Sector angles are proportional to frequency: 172.8°, 133.2° and 54°.

WORKED EXAMPLE · LUNCH CHOICES FOR 200 STUDENTS

School 96, packed 74, off-site 30.

1. School: 96/200×360 = 172.8°.
2. Packed: 74/200×360 = 133.2°.
3. Off-site: 30/200×360 = 54°.

Angles sum to 360°.

CHAPTER 10 · PRACTICE

10.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A pie chart is poor for comparing many similar categories or several groups side by side; a clustered bar chart is clearer.

GUIDED PRACTICE · FIRST STEPS

10.2.1 Find pie-chart angles for frequencies 96, 74 and 30 out of 200. FLUENCY

10.2.2 What percentage of 200 is 96? FLUENCY

10.2.3 Why should a bar-chart frequency axis start at zero? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.2.4 Year 7 has 70 students, Year 8 and Year 9 have 65 each. Which display compares lunch choices across years? APPLICATION

10.2.5 Three sector percentages are 48%, 37% and 15%. Do they fit one whole? REASONING

10.2.6 A category has sector angle 54° in a survey of 200. Find its frequency. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · MATHEMATICAL DEVELOPMENT

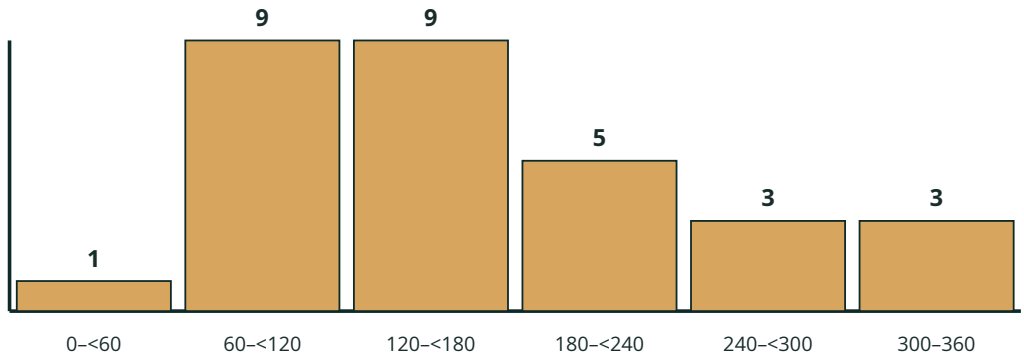
10.3 Dot plots, histograms and averages

BIG IDEA

Describe shape, centre and spread; choose an average that fits the data and the question.

A dot plot shows individual values. A histogram groups continuous or finely measured numerical data into intervals; with equal widths, bar height is frequency. State the boundary rule.

Mean = sum÷count; median is the middle ordered value (or mean of the two middle values); mode is most frequent; range = maximum−minimum. Outliers pull the mean more than the median.



Equal interval widths; heights show frequency.

Equal 60-minute intervals. Boundary rule: left endpoint included, right excluded; the final interval includes 360.

WORKED EXAMPLE · SUMMARISE 30 SCREEN-TIME VALUES

The ordered data have sum 4,740; the 15th and 16th values are 135 and 140; 120 occurs four times.

1. Mean = 4740/30 = 158 min.
2. Median = (135+140)/2 = 137.5 min.
3. Mode = 120 min; range = 360−45 = 315 min.

Mean 158; median 137.5; mode 120; range 315 minutes.

CHAPTER 10 · PRACTICE

10.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

For grouped data, a midpoint estimate of the mean is an estimate, not the exact mean of the raw values.

GUIDED PRACTICE · FIRST STEPS

10.3.1 Find the mean, median, mode and range of 2, 3, 3, 8. FLUENCY

10.3.2 For the 30 screen-time values, sum 4,740, middle values 135 and 140, mode 120, min 45, max 360. Find all four summaries. APPLICATION

10.3.3 Why is the median lower than the mean for the screen-time data? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.3.4 Group the screen data into 0–<60, 60–<120, 120–<180, 180–<240, 240–<300, 300–360 (last includes 360). Find frequencies. APPLICATION

10.3.5 Why must a histogram state its interval boundaries? REASONING

10.3.6 Estimate the mean from grouped intervals with midpoints 30, 90, 150, 210, 270, 330 and frequencies 1, 9, 9, 5, 3, 3. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · MATHEMATICAL DEVELOPMENT

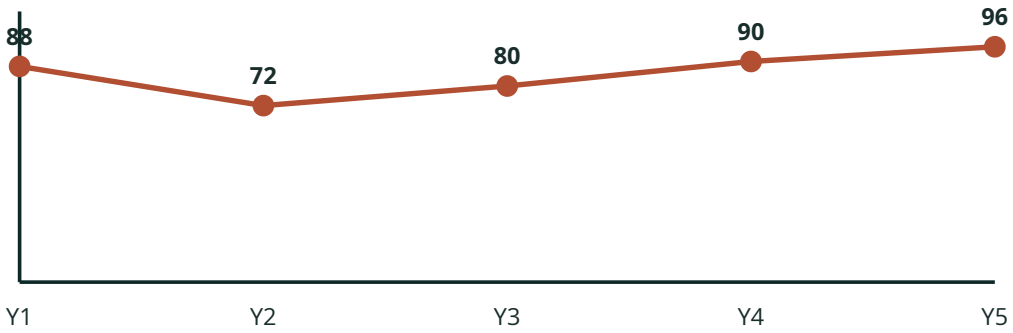
10.4 Trends, comparisons and misleading graphs

BIG IDEA

Read the scale, denominator and time order before accepting a visual claim.

A line graph or time-ordered bar chart can show change over time. Describe direction, size of change and exceptions; do not call every fluctuation a trend.

Misleading displays include truncated axes, unequal intervals, 3-D distortion, area used for a one-dimensional value, missing denominators and selective time windows. Compare like with like.



Zero-based axis; time order left to right.

A zero-based axis shows the fall from 88 to 72 and the later recovery to 96 without exaggeration.

WORKED EXAMPLE · TEST "SALES DOUBLED"

Sales rose from 50 to 90.

1. Increase = 40.
2. Percentage increase = 40/50×100 = 80%.

The claim is false: sales rose by 80%, not 100%.

CHAPTER 10 · PRACTICE

10.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A percentage can be correct while the headline is misleading if the base is tiny or the denominator is hidden.

GUIDED PRACTICE · FIRST STEPS

10.4.1 Uptake over five years is 88, 72, 80, 90, 96. Describe the overall change and the largest fall. FLUENCY

10.4.2 Find the percentage fall from 88 to 72. APPLICATION

10.4.3 A headline says “sales doubled” after 50 to 90. Test it. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

10.4.4 A pictogram doubles both height and width for a value twice as large. Explain the distortion. REASONING

10.4.5 Year 9 off-site is 13/65; Year 7 is 8/70. Compare percentages fairly. APPLICATION

10.4.6 Incidents rise from 2 to 4. Is “doubled” correct? What else should a reader ask? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 10 · EXPLORATION

How should data be grouped?

Daily screen time (minutes), ordered: 45, 60, 60, 75, 80, 90, 90, 90, 105, 110, 120, 120, 120, 120, 135, 140, 150, 150, 165, 180, 180, 195, 210, 210, 240, 240, 270, 300, 330, 360. Boundary rule: left included, right excluded; final interval includes 360.

> **INVESTIGATION**
> Criterion B. Use the same 30 values with two interval widths, compare the pictures and justify a display for families.

INTERVAL WIDTH	FREQUENCIES
60 min: 0–<60, ..., 300–360	
120 min: 0–<120, 120–<240, 240–360	

EXPLORATION · HOW SHOULD DATA BE GROUPED?

Criterion B. Use the same 30 values with two interval widths, compare the pictures and justify a display for families.

B1 Find the six 60-minute frequencies. INVESTIGATION

B2 Find the three 120-minute frequencies. INVESTIGATION

B3 Which grouping shows the central cluster more clearly? Which loses more detail? INVESTIGATION

B4 Find mean, median, mode and range. How many students spend 240 minutes or more? APPLICATION

B5 Test “the typical student spends four hours a day” and recommend a newsletter display. CHALLENGE

CHAPTER 10 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	Classify “number of siblings”.	CHECK
C2	Find the sector angle for 30 out of 200.	CHECK
C3	Find the median of 1, 3, 3, 6, 7.	CHECK
C4	A value rises from 50 to 90. Find the percentage increase.	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 10 · APPLICATION

The canteen survey

Lunch choice by year group is shown below. Prices: school lunch $4.20, packed lunch ingredients about $3.10, off-site meal about $5.50. Five-year school-lunch uptake: 88, 72, 80, 90, 96.

REAL-LIFE APPLICATION

Criterion D. Simulated whole-school survey of 200 students. State denominators and distinguish spending from canteen revenue.

YEAR	SCHOOL	PACKED	OFF-SITE	TOTAL
7	38	24	8	70
8	31	25	9	65
9	27	25	13	65
Total	96	74	30	200

APPLICATION · THE CANTEEN SURVEY

Criterion D. Simulated whole-school survey of 200 students. State denominators and distinguish spending from canteen revenue.

D1 Find the school-lunch percentage and the pie-chart angles for all three choices. APPLICATION

D2 Compare off-site eating in Year 9 and Year 7 using percentages. APPLICATION

D3 Find daily spending by lunch choice and total spending. How much extra canteen revenue if 24 students switch from packed to school lunch? APPLICATION

D4 Evaluate “off-site eating is out of control” and write a more accurate headline of at most twelve words. REASONING

CHAPTER 10 · ASSESSMENT PRACTICE

The data explainer

MYP ASSESSMENT PRACTICE

Criteria A and C. Choose a display, compute one summary and write a claim that the data actually support.

Use the lunch survey to explain why a truncated bar chart or a single pie chart could mislead.

ASSESSMENT PRACTICE · THE DATA EXPLAINER

Criteria A and C. Choose a display, compute one summary and write a claim that the data actually support.

S1 Which display best compares lunch choices across the three year groups? Explain. ASSESSMENT

S2 Calculate the school-lunch share and one check on the pie chart. ASSESSMENT

S3 Write one supported claim and one warning about a misleading display. ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 10 · MIXED PRACTICE

Mixed practice 10

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 10

Connect ideas. Give units and reasons where requested.

M1	Classify “height in centimetres”.	FLUENCY
M2	Find the sector angle for 74 out of 200.	FLUENCY
M3	Find mean, median, mode and range of 4, 4, 6, 10.	FLUENCY
M4	A frequency table has counts 12, 18, 20. Find the percentage for 18.	APPLICATION
M5	Why can a truncated vertical axis mislead?	REASONING
M6	Uptake falls from 90 to 72. Find the percentage fall.	APPLICATION
M7	A data set has mean 50 and median 30. What might explain the gap?	REASONING
M8	Grouped intervals 0–<10 and 10–20 have frequencies 4 and 6. Estimate the mean using midpoints.	CHALLENGE
M9	A survey of 80 has 20 choosing A. Find the percentage and pie angle.	APPLICATION
M10	A headline says “crime doubled” from 2 to 4 incidents. Give a balanced response.	CHALLENGE

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 10 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **200.**
Add the category counts.

R2 **48%.**
Divide and multiply by 100.

R3 **4.**
Sum 12÷3.

R4 **90°.**
A quarter turn.

R5 **2, 3, 5, 8.**
Sort before finding a median.

CHECKPOINT

C1 **Discrete numerical.**
It counts separate whole values.

C2 **54°.**
30/200×360.

C3 **3.**
The middle ordered value.

C4 **80%.**
40/50×100.

CHAPTER 10 · ANSWERS AND REASONING

Answers · 10.1 and 10.2

PRACTICE 10.1 · DATA AND VARIABLES

10.1.1 **Nominal categorical; ordinal categorical; discrete numerical; continuous numerical.**
Name, order, count and measurement are different variable types.

10.1.2 **96.**
Add the three counts.

10.1.3 **It is an ordered category or label; an average year code may not describe a meaningful category.**
Variable type determines appropriate summaries.

10.1.4 **70; 38/70 ≈ 54.3%.**
Row total 70; use the row as denominator.

10.1.5 **No. They are category labels, not counts or measurements.**
Nominal codes have no numerical order or distance.

10.1.6 **30; 15%.**
200−96−74; 30/200×100.

PRACTICE 10.2 · BAR AND PIE CHARTS

10.2.1 **172.8°, 133.2°, 54°.**
Multiply each share by 360°.

10.2.2 **48%.**
96/200×100.

10.2.3 **Bar height encodes frequency; a truncated axis exaggerates differences.**
Start the frequency axis at zero so the visual ratio matches the data ratio.

10.2.4 **A clustered bar chart (or a well-labelled two-way table).**
It keeps year groups visible side by side.

10.2.5 **Yes, they sum to 100%.**
A pie chart partitions the whole.

10.2.6 **30.**
54/360×200.

CHAPTER 10 · ANSWERS AND REASONING

Answers · 10.3 and 10.4

PRACTICE 10.3 · DISTRIBUTION AND AVERAGES

10.3.1 **Mean 4; median 3; mode 3; range 6.**
Sum 16÷4; middle two 3 and 3; 8−2.

10.3.2 **Mean 158; median 137.5; mode 120; range 315 minutes.**
Use the definitions directly.

10.3.3 **A tail of high values pulls the mean upward; the median depends only on position.**
The distribution is right-skewed.

10.3.4 **1, 9, 9, 5, 3, 3.**
Count each value once using the stated boundary rule.

10.3.5 **A value on a boundary must belong to exactly one interval; otherwise counts can be duplicated or omitted.**
Equal widths also make heights comparable.

10.3.6 **168 minutes.**
Weighted midpoint sum ÷30 = 168. A grouped estimate need not equal the raw mean (158 minutes).

PRACTICE 10.4 · TRENDS AND CLAIMS

10.4.1 **Overall +8; largest fall 16 from year 1 to year 2.**
96−88=8; 88−72=16.

10.4.2 **About 18.2%.**
16/88×100.

10.4.3 **False: 80% increase, not 100%.**
Doubling would require 100.

10.4.4 **The area becomes four times as large, exaggerating a twofold value.**
Area scales with the square of linear size.

10.4.5 **20% versus about 11.4%; Year 9 is higher.**
Use each year group as its denominator.

10.4.6 **Yes, 100% increase, but ask for the time period, population size and longer trend; small counts can make a dramatic headline.**
A relative change needs context.

CHAPTER 10 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · HOW SHOULD DATA BE GROUPED?

B1	**1, 9, 9, 5, 3, 3.** Count each value once.
B2	**10, 14, 6.** Combine adjacent 60-minute intervals.
B3	**60-minute intervals show the cluster around 60–180 more clearly; 120-minute intervals lose more detail.** Narrower equal intervals preserve more shape.
B4	**Mean 158; median 137.5; mode 120; range 315 minutes. Six students are at 240 or more.** Use the raw ordered data; count 240,240,270,300,330,360.
B5	**Not supported: mean 158 min, median 137.5 min; only 6/30=20% reach 240 min. Recommend a labelled 60-minute histogram with median and a note about the high-value tail.** Choose an average and grouping that match the claim and audience.

CHAPTER 10 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE CANTEEN SURVEY

D1 **48%; angles 172.8°, 133.2°, 54°.**
Use denominator 200.

D2 **Year 9: 20%; Year 7: about 11.4%.**
13/65 and 8/70.

D3 **School $403.20; packed $229.40; off-site $165.00; total $797.60. Extra canteen revenue $100.80.**
Multiply counts by prices; the switch adds 24×4.20 to canteen revenue.

D4 **Sample headline: "Off-site lunches are 15% overall, higher in Year 9." The claim is not supported by one share alone; compare year groups and time data.**
15% overall and 20% in Year 9 are descriptive, not proof of a crisis.

ASSESSMENT PRACTICE · THE DATA EXPLAINER

S1 **A clustered bar chart or two-way table; it preserves year-group comparisons.**
A single pie chart hides the year structure.

S2 **48%; angles sum to 360° (172.8+133.2+54).**
The whole is 200 students.

S3 **Sample claim: "School lunch is the most common choice (48%)." Warning: a bar axis not starting at zero can exaggerate differences.**
The claim quotes a denominator and the warning names a visual distortion.

CHAPTER 10 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 10

M1 **Continuous numerical.**
It is measured on a scale.

M2 **133.2°.**
74/200×360.

M3 **Mean 6; median 5; mode 4; range 6.**
Sum 24÷4; middle (4+6)/2; 10−4.

M4 **36%.**
18/50×100.

M5 **It exaggerates visual differences by hiding the zero baseline.**
Bar height should be proportional to frequency.

M6 **20%.**
18/90×100.

M7 **A high-value tail or outliers pulling the mean upward.**
The median is resistant to extreme values.

M8 **11.**
(4×5+6×15)/10.

M9 **25%; 90°.**
20/80×100 and ×360.

M10 **The 100% increase is correct, but the base is tiny; ask for population, time period and longer trend before judging risk.**
Relative change without context can mislead.